

Problem 1 (Putnam 2008)

We claim the answer is, $(a \ominus b) = f(a) - f(b)$ for any real function f . This obviously works.

Plugging in $a = b = c$, we get, $(a \ominus a) = 0$ for all real a .

Plugging in $b = c$, we get, $(a \ominus b) = -(b \ominus a)$. Now, plug in $c = 3.14$ to get,

$$(a \ominus b) + (b \ominus 3.14) + (3.14 \ominus a) = 0$$

$$\Leftrightarrow (a \ominus b) = (a \ominus 3.14) - (b \ominus 3.14)$$

Note that we can write $f(x) = (x \ominus 3.14)$ for some real function f to get desired.

Problem 2 (Romania TST)

We claim the only solutions are $m = n = 1$ for any positive integer x, y or $x = y = 2^{\frac{n-1}{n-m}}$ for any positive integers m, n that leaves a positive integer x, y .

Over positive integers, we clearly have $m \leq n$ since otherwise $LHS \geq x^m + y^m > x^n + y^n = RHS$.

Case 1: $m = n$

We have $(x + y)^m = x^m + y^m$. By binomial theorem, this is only possible if $m = n = 1$.

Case 2: $m < n$

We let $d = \gcd(x, y)$. We now write $a = \frac{x}{d}; b = \frac{y}{d}$ such that a, b are relatively prime.

$$(a + b)^m = d^{n-m}(a^n + b^n)$$

Subcase 2a: $a = 2; b = 1; n = 3$

We get, $3^m = 9 \cdot d^{n-m} \Leftrightarrow m = 2 + (3 - m) \log_3 d$ where $k = \log_3 d$ must be a non-negative integer. There is no solution for $m = 3$, while we have $k < 0$ if $m > 3$ or $m < 2$. The only solution is hence $m = 2, d = 1$.

This yields the edge case $x = 2; y = 1; m = 2; n = 3$.

Subcase 2b: $a = b$

This means $x = y$ as well, so,

$$(2x)^m = x^n + x^n$$

$$\Leftrightarrow 2^m \cdot x^m = 2 \cdot x^n$$

$$\Leftrightarrow 2^{m-1} = x^{n-m}$$

$$\Leftrightarrow m - 1 = (n - m)(\log_2(x))$$

This gives us another family of solutions $x = y = 2^{\frac{n-1}{n-m}}$.

Subcase 2c: All other cases for $a \neq b$. WLOG, let $a > b \geq 1$. We don't have the case where $a = 2; b = 1; n = 3$, and $\gcd(a, b) \neq 1$, so we can use Zsigmondy's Theorem.

There exists a prime p such that $p \mid x^n + y^n$ but $p \nmid x + y \Rightarrow p \nmid (x + y)^m$. Therefore, the equation is impossible to satisfy since the RHS is divisible by p and the LHS isn't.

14 clubs; 3.5 hours